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The definition and measurement of engagement and adherence for digital mental health interventions: an umbrella review

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ABSTRACT

Digital Mental Health Interventions (DMHIs) are scalable and accessible, but low engagement and adherence limit their effectiveness. Varying definitions and measures of engagement and adherence hinder effective evaluation and comparison. This umbrella review synthesised 61 systematic reviews, meta-analyses, and scoping reviews, which focused on DMHIs and provided a definition or measurement of engagement and/or adherence. Inclusion and exclusion were guided by PICOS, with record counts reported in the PRISMA flowchart. Results showed that adherence and engagement were often undefined. When defined, engagement was most often combined with adherence or described as extent of programme usage, with completion metrics and usage metrics being the two most common measurements. When defined, adherence was most often described as programme/module/session completion or combined with engagement, with completion metrics or a combination of completion and usage metrics being the two most common measurements. Factors influencing engagement and adherence clustered into three themes: (i) participant level factors (e.g. demographic variables), (ii) technology level factors (e.g. privacy concerns), and (iii) programme level factors (e.g. perceived usefulness of the DMHI). Results are discussed in terms of how engagement and adherence to DMHIs are defined, measured, and promoted, highlighting priorities, challenges and recommended practices.

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review; digital health

1. Introduction

Digital mental health interventions (DMHIs) leverage technology to provide psychological support, information, and therapy through platforms such as mobile apps, web-based programmes, and telehealth services. Designed to enhance mental well-being and assist individuals in managing psychological conditions (Brotherdale et al. 2024; Williams et al. 2021), these interventions offer flexible, affordable, and accessible alternatives to traditional in-person care, especially for marginalised populations with limited access to face-to-face services (Bond et al. 2023; Zhou et al. 2020). With the global need for scalable mental health solutions on the rise, DMHIs are increasingly recognised as integral components of modern public health initiatives.

Despite their potential, DMHIs often exhibit low engagement and poor adherence, limiting their long-term effectiveness and scalability (Baumel et al. 2019; Lipschitz et al. 2023). Many users discontinue use prematurely or fail to engage with the content in a sustained and meaningful manner. A primary obstacle to

addressing these challenges is the absence of consensus regarding the definitions and measurement of engagement and adherence in the literature.

Engagement with digital mental health interventions (DMHIs) has been assessed in varied ways across the literature. Some studies rely on narrow behavioural indicators, such as frequency of logins or total usage time (Boucher and Raiker 2024), whereas others adopt broader measures, including uptake, sustained participation, and completion rates (Lipschitz et al. 2023). In certain cases, more superficial metrics like the number of downloads are inaccurately treated as evidence of meaningful engagement, despite providing limited information about the user experience or therapeutic outcomes (Hyzy et al. 2024). Increasingly, scholars have called for a more nuanced understanding of engagement that incorporates cognitive factors (e.g. attention, understanding) and emotional aspects (e.g. motivation, satisfaction) in addition to behavioural measures. Such a multidimensional approach highlights the importance of quality of interaction with DMHIs, rather than

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focusing exclusively on usage quantity (Hoffman et al. 2023; Nahum-Shani et al. 2022; Ramadurai et al. 2022).

Similarly, adherence is often measured inconsistently across DMHI studies. It is frequently treated as a binary outcome ('completed versus not completed'), without capturing variations in how users interact with the intervention or the quality of their participation (Aschbacher et al. 2023; Forbes et al. 2023; Valentine et al. 2025). Emerging approaches highlight the importance of evaluating both the quantity and appropriateness of use, including whether users meet a minimum effective dose or follow intended instructions (Smith, Ward, and Lambe 2025). Both engagement and adherence also overlap with related concepts such as completion, retention, and attrition, further complicating consistent measurement and reporting (Borghouts et al. 2021).

Existing definitions and methods for measuring digital mental health interventions vary widely. This lack of consensus complicates the comparison of research findings, impedes the establishment of standardised evaluation frameworks, and hinders the development of effective, evidence-based approaches to enhance user engagement and retention. Resolving these issues is critical to improving the design, implementation, and assessment of DMHIs. In the absence of clear conceptual and methodological guidance, it becomes challenging to identify which interventions are most effective, for which populations, and under what circumstances.

1.1. Research aims

This umbrella review synthesises evidence from systematic reviews, meta-analyses, and scoping reviews to provide a comprehensive overview of how engagement and adherence are conceptualised, measured, and influenced in the context of digital mental health interventions (DMHIs). Guided by the Population, Concept, and Context (PCC) framework recommended by the Joanna Briggs Institute (Aromataris and Munn 2020), and as outlined in the published protocol (Yap and Lin 2025), the review addresses the following questions:

- How is engagement defined and measured in systematic reviews, meta-analysis, and scoping reviews of Digital Mental Health Interventions?
- How is adherence defined and measured in systematic reviews, meta-analysis, and scoping reviews of Digital Mental Health Interventions?
- What are the factors considered to influence DMHI Engagement and Adherence?

2. Methods

This review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al. 2021) and its extension for umbrella reviews (Li et al. 2021). The completed PRISMA checklist is available in the published protocol (Yap and Lin 2025). The review protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO; CRD42025637603) and published in *JMIR Research Protocols* (Yap and Lin 2025), available at <https://doi.org/10.2196/73438>.

2.1. Search strategy

A systematic search was conducted on 5 March 2025, across Web of Science, CINAHL, MEDLINE, PsycINFO, Cochrane via EBM Reviews, and ProQuest Complete. BMC Digital Health was also hand searched to identify relevant studies not yet indexed in major databases. These databases were selected for their comprehensive coverage of peer-reviewed literature in psychology, healthcare, and digital health interventions, consistent with the methodologies employed in comparable systematic reviews (Bear et al. 2024; Elkes et al. 2024).

Five key publications (Bear et al. 2024; Boucher and Raiker 2024; Elkes et al. 2024; Forbes et al. 2023; Lipschitz et al. 2023) were selected by the authors based on their relevance to the inclusion criteria; this process is further elaborated in the pre-registered protocol (Yap et al., 2025). The sensitivity of CINAHL and Cochrane was evaluated by determining the proportion of these key studies each database identified (McGinn et al. 2016). CINAHL provided several unique relevant records not available in other databases, supporting its inclusion for comprehensive coverage. Cochrane was included due to its recognised role as a specialised source for high-quality systematic reviews and evidence syntheses (Puga and Atallah 2020).

The searches targeted key concepts related to *engagement* and *adherence* within digital contexts (e.g. web-based, mobile, app-based, and computer-delivered platforms) for *mental health* and *well-being* (including emotional health, anxiety, and stress) as well as *interventions* spanning promotion, prevention, and treatment. The search strategy was designed in consultation with a subject librarian and underwent validation procedures (Aromataris and Riitano 2014). Because of matters such as inconsistent terminology and a rapidly developing area, forward citation chaining was used to capture studies potentially missed due to indexing gaps or limited database coverage. In line with recommendations from the *Cochrane Handbook* (Lefebvre, Glanville, and

Briscoe 2025), the research team assessed the strategy's robustness by verifying that it successfully retrieved a set of known, relevant publications.

The research team identified peer-reviewed systematic reviews, scoping reviews, and meta-analysis published between January 2015 and March 2025 in English. Gray literature was excluded, but the citationchaser (Haddaway, Grainger, and Gray 2022) tool was utilised to conduct supplementary searching via forward citation-chaining. Please refer to Table 1 for the line-by-line search strategy. The full search strategy, including database-specific operators and the manual handsearching, is available in the published protocol (Yap et al., 2025).

2.2. Eligibility criteria

The Population, Intervention, Comparator, Outcome, Study Design (PICOS) framework was applied to guide the inclusion and exclusion of studies. The complete PICOS table is available in the published protocol (Yap and Lin 2025). Included studies focused on digital mental health interventions (DMHIs) that provided a definition or measurement of engagement and/or adherence. Eligible study designs comprised systematic reviews, meta-analyses, and scoping reviews that conducted a systematic search of at least one database and followed PRISMA guidelines. Systematic reviews that included other study types (e.g. qualitative studies, case studies) were also eligible, but analysis was focused on the systematic review component wherever possible.

Table 1. Line-by-line search strategy (Yap and Lin 2025).

Line	Search Terms
1	Engag* OR Participat* OR Interact* OR Involv* OR Adopt* OR Use* OR Usage OR "Log In" OR "Sign In" OR Download*
2	Adher* OR Compliance OR Compliant OR Complying OR Complied OR Complete OR Completion OR Finish OR Retention OR Retain* OR Commit*
3	1 or 2
4	digital* or mHealth or "m-health" or eHealth or "e-health" or "web-based" or "internet-based" or "mobile phone*" or "cell* phone*" or "smart phone*" or smartphone* or "mobile device*" or "text messag*" or "text-based" or SMS or app* or "tele-health" or telehealth or computer* or online or "electronic health" or telemedicine or "tele-medicine"
5	"mental health" or "well-being" or wellbeing or "psychological health" or "emotional health" or "behavioral health" or "mental difficult*" or "mental illness*" or "mental disorder*" or "mental disease*" or "mood disorder*" or depress* or anxiet* or anxious or suicid* or stress* or psychotherap* or psychiatr*
6	Interven* or promot* or prevent* or program* or support* or policy or policies or implement* or evaluat* or therap* or develop* or Treatment* or help*
7	where possible, combine 4 and 5 and 6 using proximity operators
8	review* or meta-analys* or metanalys* or meta analys*
9	3 and 7 and 8
10	Limit 9 to English language and yr = "2015-Current") and Peer-Reviewed Journals

Note that in cases of screening preprint articles, efforts were undertaken to see if they had been peer-reviewed and updated since. Published versions (and their updated details) were recorded and considered for review.

Studies were required to (1) report a direct mental health outcome, as defined by the *Diagnostic and Statistical Manual of Mental Disorders* (DSM) linked to an intervention (i.e. not limited to mental health support, promotion, or literacy), to ensure that the intervention addressed a mental health condition rather than a mental health-adjacent construct (e.g. enabling the exclusion of studies focused on bullying, help-seeking, or quality of life, etc.) and (2) include a definition and/or measurement of engagement and adherence, or similar metrics such as attrition, retention, or uptake, provided these were discussed beyond simple data extraction and in conjunction with, or as proxies for, engagement or adherence.

Exclusion criteria were applied to studies that examined digital interventions without a mental health outcome, addressed non-digital mental health interventions, or discussed digital mental health in the absence of an intervention component. Reviews were also excluded if they did not adequately define or measure engagement and adherence. In addition, ineligible study types (such as narrative reviews, other umbrella reviews, primary research, or protocols) were omitted. No restrictions were placed on comparators or study populations.

2.3. Data extraction

Data extraction was carried out using Covidence's Extraction 2 tool (Covidence Systematic Review Software n.d.). Extracted information included bibliographic details (authors, year), review type (systematic review, meta-analysis, or scoping review), study objectives, synthesis methods, number and type of included studies, participant characteristics, descriptions of the digital mental health interventions, outcome measures, and key findings. Extraction was guided by the JBI Data Extraction Form for Systematic Reviews and Research Syntheses (Aromataris and Munn 2020). To assess the quality of the primary data, overall critical appraisal scores and quality assessment indicators were also extracted.

The extracted information was summarised in tables detailing review characteristics, characteristics of included studies, participant profiles, and study outcomes. In cases where no information was available on the date range of the search, attempts were made to locate associated research protocols, supplementary

materials, or to infer the timeframe from the list of included studies. For the population section, settings and context were only specified if relevant to the PICOS framework (i.e. if the research question specified these parameters). Otherwise, general details about the population were extracted.

Where the total number of participants was not reported, counts were manually calculated from the table of included study characteristics when available, except for variable real-world usage (Fleming et al. 2018). Data extraction was performed by the primary reviewer (LKY), with a random 10% checked by a second reviewer (EE). Discrepancies were discussed and resolved, with additional checks conducted until consensus was reached.

2.4. Quality assessment

Critical appraisal was conducted using Covidence's Extraction 2 tool (Covidence Systematic Review Software n.d.). The methodological quality of the included reviews and meta-analyses was evaluated using the JBI Critical Appraisal Checklist for Systematic Reviews and Research Syntheses (Joanna Briggs Institute, 2017), which consists of 11 items assessing aspects such as the clarity of the review question, appropriateness of inclusion criteria, adequacy of the search strategy, methods for critical appraisal, and the validity of data synthesis and conclusions.

For the within-review quality rating (the quality of the included studies within each review) assessments were made based on the judgement of the majority (more than half) of the included studies. Quality assessment was performed by the primary reviewer (LKY), with a random 10% checked by a second reviewer (EE). Discrepancies were discussed and resolved, with additional checks conducted until consensus was reached. Studies were not excluded on the basis of quality. Both the JBI critical appraisal results and the overall risk of bias for each review's included studies were reported.

2.5. Data synthesis

Definitions and measurements of engagement and adherence were synthesised using a JBI-style meta-aggregation approach. This assembles evidence without reinterpreting the original authors' findings (Hannes and Lockwood 2011; Lockwood et al. 2020). The convergent mixed-methods approach described in the protocol could not be implemented due to a lack of sufficient quantitative data. Although not all included studies were qualitative, meta-aggregation was

considered suitable for the analysis because it enabled a synthesis focused on the conceptualisation and operationalisation of engagement and adherence, both of which are rooted in qualitative evidence.

First, each study's reported definition or measurement was extracted as a separate finding, retaining illustrative quotes exactly as reported. Next, findings that were conceptually similar were grouped into categories, and wherever possible these categories were further combined according to shared meaning to create overarching synthesised statements, providing a systematic and transparent summary of the conceptual evidence. Finally, the frequency with which each approach appeared across studies was recorded to indicate relative prominence, highlighting both common and less frequently reported methods. The meta-aggregation process was led by the primary author, who independently developed the categories and synthesised statements. Full meta-aggregated data are available in Supplementary Materials 1 and 2 for engagement and adherence, respectively.

Factors influencing engagement and adherence were synthesised using a thematic synthesis approach (Thomas and Harden 2008). First, quotes describing determinants of engagement were extracted and coded inductively. Next, these quotes were organised into overarching themes at the participant, technology, and programme levels. This organisation enabled interpretive analysis of qualitative insights, revealing patterns and broader contextual influences on engagement and adherence. The results were then narratively synthesised to inform recommendations for co-production, inclusivity, and implementation of digital mental health interventions. Full coding and thematic development are provided in Supplementary Material 3 to ensure transparency of the analytic process.

3. Results

3.1. Study selection

The search initially yielded 5,087 studies. An additional 35 records were identified through manual handsearching of *BMC Digital Health*, and 4,515 more were identified through forward citation chasing. After removing duplicates, with 3,120 automatically identified by Covidence and 1,094 manually identified, 5,423 studies underwent title and abstract screening. A moderate level of agreement ($\kappa = 0.59$) was achieved between the two independent reviewers (LY and EE), with conflicts resolved by a third reviewer (MM).

A total of 225 articles were deemed potentially eligible and retrieved for full-text screening. The primary reviewer (LY) conducted full-text screening, which

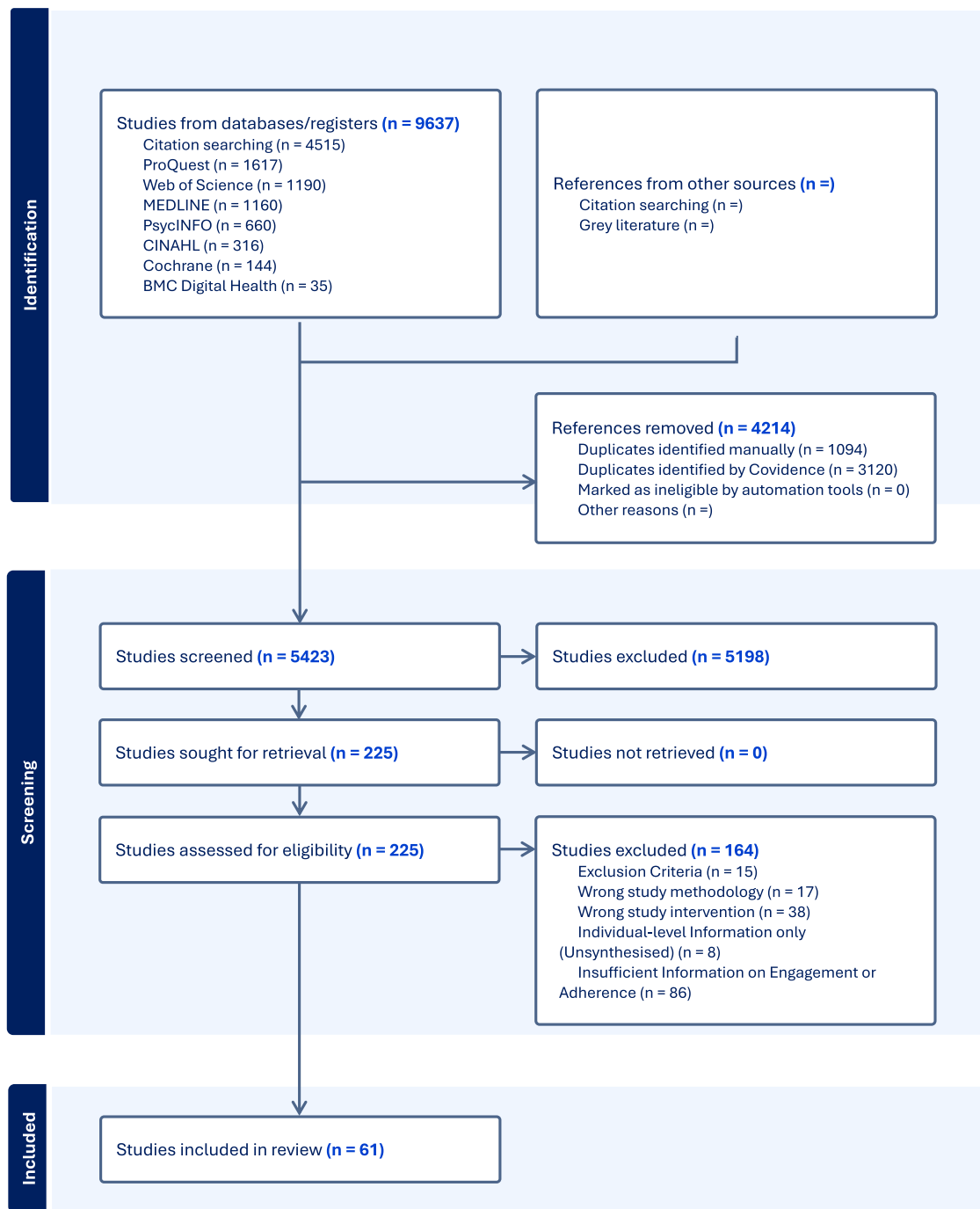


Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Flowchart from Covidence.

was checked by a second reviewer (EE). This stage yielded almost perfect agreement ($\kappa = 0.97$). Of the 225 articles, 15 were excluded based on exclusion criteria, 17 for having an ineligible study methodology (e.g. reviews that did not conduct a systematic search or follow PRISMA guidelines), 38 for evaluating the wrong type of intervention (interventions that don't affect direct mental health outcomes e.g. adaptive strategies or user acceptance), 86 for providing insufficient information on engagement and adherence, and eight

for including only individual study-level data without discussing a definition or measurement.

This process resulted in 61 articles being included in the review. Figure 1 presents the PRISMA flowchart. Of these, 25 (41.0%) were systematic reviews, 21 (34.4%) were systematic reviews with meta-analyses, 10 (16.4%) were scoping reviews, and four (6.6%) were meta-analyses alone. One integrative review (1.6%) was also included because it did not meet exclusion criteria and addressed the research questions.

3.1.1 Characteristics of included studies

A total of 61 reviews, including 2,876 studies, across 318,365 participants (reported where participant total was available or could be inferred from the table/protocol) were included in this umbrella review. The study characteristics are summarised in Supplementary Material 4. The earliest date of search is database inception, as indexes were available as early as 1946 for MEDLINE, 1806 from PsycINFO, and 1947 for EmBASE (Gan et al. 2021). Among articles that did not go as far as database inception, the earliest search ranged from January 1990. Publication rates increased markedly following the onset of the COVID-19 pandemic, with 68.9% of studies published between 2020 and 2024. The highest numbers appeared in 2022 ($n = 15$), followed by 2021 and 2023 ($n = 9$ each). Prior to 2020, fewer than five studies were published annually.

The 61 reviews originated from 20 countries, predominantly in Oceania, Europe, and North America. Australia contributed the most ($n = 18$), followed by the United Kingdom ($n = 12$) and the United States ($n = 9$). Other countries included Canada ($n = 4$), Germany, and the Netherlands ($n = 2$ each), and 14 countries with a single study each. By continent, studies were mostly from Europe ($n = 26$), Oceania ($n = 19$), and North America ($n = 13$), with fewer from Asia ($n = 3$).

Populations studied varied, with most focusing on clinical samples ($n = 21$) or no specific population restrictions ($n = 19$). Clinical samples primarily targeted depression and anxiety, alongside bipolar disorder, PTSD, OCD, insomnia, and psychosis. Young people were the focus of 10 studies, with fewer targeting adults only ($n = 3$), students ($n = 3$), or racial/ethnic minorities ($n = 2$). Unique populations studied included men who have sex with men (Clark and Kim 2021), perinatal individuals (Davis et al. 2022), and familial or relational dyads (Welsh et al. 2024).

Depression and anxiety, individually or combined, were the most common conditions targeted (84%), followed by stress (18%) and general mental well-being (8%). Eating disorders, substance abuse, and PTSD appeared in about 7% of studies each, while psychosis, bipolar disorder, and OCD were addressed in roughly 5%. Insomnia, quality of life, ADHD, personality disorders, and broader constructs such as mindfulness and social connectedness were less frequently examined.

Digital delivery formats varied widely, reflecting technological evolution. Mobile applications were most common (30%), followed by web-based platforms (20%) and internet-based cognitive behavioural therapy (7%). Other formats included online programmes (10%), blended or videoconference interventions, ecological momentary assessment/intervention, and

various less conventional methods (20%), such as CD-ROMs (Drews-Windeck, Greenwood, and Cavanagh 2023; Forbes et al. 2023; Li et al. 2022; Linardon and Fuller-Tyszkiewicz 2020), social networking services (Ridout and Campbell 2018), bulletin boards (Biagianti, Quraishi, and Schlosser 2018), wearable technology (Borghouts et al. 2021; Oti and Pitt 2021), blogging platforms (Babbage et al. 2022), virtual communities (Drews-Windeck, Greenwood, and Cavanagh 2023), fitness trackers (Raeside et al. 2024), avatar-based programmes (Ehrt-Schäfer et al. 2023), and downloadable text or audio content (Polak and Tanzer 2024).

3.2. Quality assessment

Overall, the quality of the included reviews was generally high, with approximately half meeting all but three or fewer items on the critical appraisal checklist. However, several methodological weaknesses were commonly observed. The most frequent issues related to whether the sources and resources used in the literature search were adequate, and whether publication bias was assessed. Alarming, only 20% of reviews examined the likelihood of publication bias. See Supplementary Material 5 for the full detailed quality assessment matrix.

Search strategies were often insufficiently justified, with about one-third of reviews failing to explain database selection, excluding grey literature without rationale, or omitting supplementary searches. Data extraction procedures were also frequently unclear or conducted by a single reviewer in 44% of cases, increasing risk of bias. Conversely, most reviews clearly reported review questions, inclusion criteria, and search strategies, reflecting protocol registration standards. Recommendations for policy and practice were generally well supported.

Critical appraisal methods varied among the included reviews. The Cochrane Collaboration Risk of Bias Tool was the most commonly used (31%), followed by the Mixed Methods Appraisal Tool (15%). Less frequently used tools included the JBI Critical Appraisal Checklist, CONSORT Checklist, and CDC Compendium (1–2% each). About 11% of reviews employed multiple tools, 5% adapted criteria from prior studies, and 3% developed custom appraisal tools. Notably, 30% of reviews did not conduct formal quality appraisals.

The quality of studies included within these reviews also varied widely. About a quarter (26%) of reviews rated the majority of included studies as high quality, indicating low risk of bias. In contrast, 18% rated most studies as low quality, suggesting higher risk of bias. Around 15% reported mixed quality levels across included studies, and

Conceptualisations of Engagement (by no. of studies)

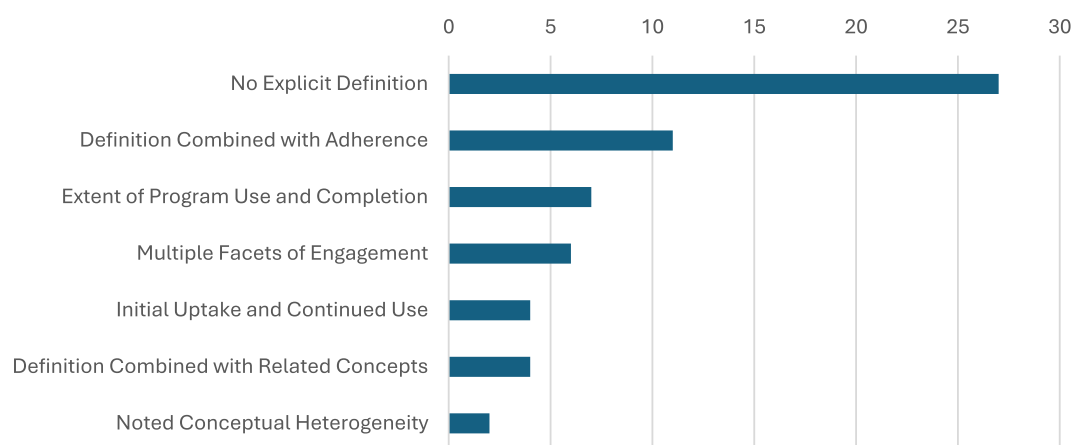


Figure 2. Summary of engagement definitions by number of studies.

8% provided unclear or incomplete quality information. Notably, a third (33%) of the reviews did not conduct a formal critical appraisal of included studies and were thus classified as 'Not Applicable' (NA).

3.2.1 Definitions of engagement

Figure 2 summarises the number of studies reporting each conceptualisation of engagement. Conceptualisation refers to how engagement was defined in each study, and was derived inductively from definitions of engagement extracted from the included studies via the meta-aggregation grouping process. The full meta-aggregation tables are illustrated in Supplementary Material 1 and 2. The most common position was that almost half of the reviews (27/61, 44.3%) did not provide an explicit definition of engagement. Secondly, in 11 reviews (18.0%), engagement was not separated from the concept of adherence.

A smaller proportion of studies offered specific conceptualisations. Seven reviews (11.5%) defined it in terms of programme usage and completion, referring to the extent to which users interacted with the intervention. Six reviews (9.8%) referred to multiple facets of engagement, incorporating cognitive, emotional, affective, and subjective experiences. Four reviews (6.6%) described engagement as a process of initial uptake (signing up/adopting the intervention) followed by continued use (sustained interaction over time) of the intervention. Reflecting definition combined with other related concepts, in four reviews (6.6%), it was not separated from other related outcomes, with engagement used interchangeably with or as part of retention (Babbage et al. 2022), uptake and dropout (Bisby et al. 2022), retention (Linardon and Fuller-Tyszkiewicz 2020), or measurement-related definitions (Ng

et al. 2019). Two reviews (3.3%) acknowledged conceptual heterogeneity, noting that engagement was defined idiosyncratically by individual authors without a standardised framework. Table 2 provides illustrative examples for each category shown in Figure 2.

3.3. Definitions of adherence

The number of studies referencing each adherence conceptualisation is shown in Figure 3. More than half (32/

Table 2. Sample text for defining engagement across included reviews.

Conceptualisations of engagement	Example definition
Definition combined with adherence	"Adherence has been defined as the extent to which individuals experience or engage with the content of the treatment." (Børtveit et al. 2022)
Extent of programme use and completion	"... engaging at the recommended frequency or complete the full course of the intervention." (Bear et al. 2024)
Multiple facets of engagement	"... a subjective experience or as a behaviour. Includes both the extent (e.g. amount, frequency, duration, and depth) of usage and the subjective experience of what it feels to be engaged (e.g. attention, interest, and affect)." (Liverpool et al. 2020)
Initial uptake and continued use	"... user's uptake and sustained interactions with a digital intervention." (Borghouts et al. 2021)
Definition combined with related concepts (<i>Retention</i>)	"... the results seem to support the notion that intervention and trial retention follow a similar relationship in terms of indicating the level of young people's engagement." (Babbage et al. 2022)
Noted conceptual heterogeneity	"... therefore there was not a clear definition of what engagement is and how it impacted results." (Antezana et al. 2022)

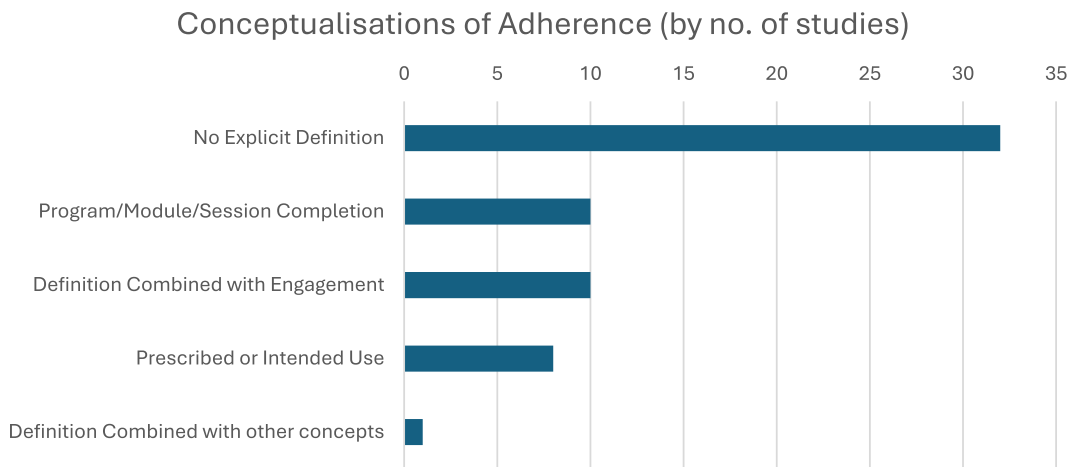


Figure 3. Summary of adherence definitions by number of studies.

61, 52.5%) did not provide an explicit definition of adherence. In terms of how adherence was defined, ten reviews defined it as completion (16.4%) referring to the completion of a programme, module, sessions, treatment, or minimum dosage, while another ten reviews (16.4%) did not separate adherence from engagement. Eight studies defined adherence according to recommended use (13.1%), which involves using the DMHI as it was prescribed or intended. One review (1.6%) combined adherence with other concepts, specifically as a proxy for usage (Liverpool et al. 2020). Sample text used for each category of adherence definitions is shown in Table 3.

3.4. Measurements of engagement

Figure 4 summarises the number of studies reporting each operationalisation of engagement. Operationalisation refers to how each study measures engagement and

Table 3. Sample text for defining adherence across included reviews.

Conceptualisations of adherence	Example definition
Programme/module/session completion	"... the proportion of participants that either fully completed the intervention or completed a defined minimum dose of the intervention." (Leung et al. 2022)
Definition combined with engagement	"Adherence: defined as the indicator of the extent to which participants used or engaged with the smartphone intervention." (Linardon and Fuller-Tyszkiewicz 2020)
Prescribed or intended use	"... the extent to which individuals' intervention use corresponds with creators' recommendations, instructions, and expectations of appropriate use." (Li et al. 2022)
Definition combined with related concepts (<i>usage</i>)	"... extent of <i>usage</i> of the DHI over time or adherence" (Liverpool et al. 2020)

is consistent with the inductive approach where measurements of engagement were extracted and synthesised using meta-aggregation. The full meta-aggregation tables are presented in Supplementary Materials 1 and 2. Measurement patterns mirrored the variation in definitions. A substantial proportion of reviews (24/61, 39.3%) did not report any measurement of engagement. Eight reviews (13.1%) assessed engagement through module completion rates or averages, calculating either the proportion or mean number of modules completed. Another eight (13.1%) relied on objective usage metrics such as login counts or time spent on the platform. Seven reviews (11.5%) combined objective and subjective approaches, pairing usage analytics with self-report measures such as the User Engagement Scale or System Usability Scale, and qualitative interviews. No reviews used subjective measures alone. Seven reviews (11.5%) used study-specific measurement approaches tailored to their design, theoretical framework, or research objectives. Four reviews (6.6%) did not separate engagement measurement from adherence, and three (4.9%) measured it in conjunction with other outcomes such as usability, satisfaction, acceptability, feasibility (Anmella et al. 2022; Bijkerk et al. 2023), and dropout (Limpanopparat, Gibson, and Harris 2024). Sample text used for each measurement category is presented in Table 4.

3.5. Measurements of adherence

The number of studies referencing each adherence operationalisation is shown in Figure 5. Similar to engagement, a good portion (26/61, 42.6%) did not report any measurement. About a third of the reviews (20/61, 32.8%) measured adherence through completion, applying similar criteria as in the definitions

Operationalisations of Engagement (by no. of studies)

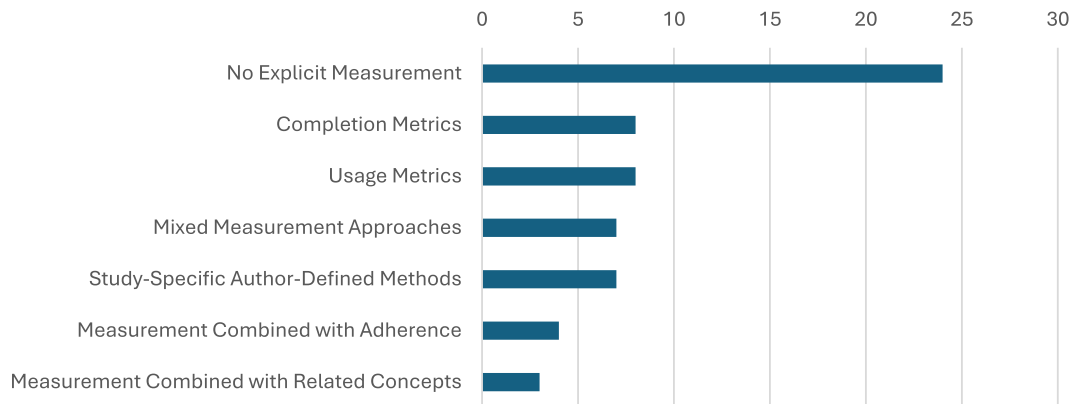


Figure 4. Summary of engagement measurements by number of studies.

(e.g. programme or session completion, achieving a minimum dosage). Seven reviews (11.5%) combined completion data with usage log information, for example reporting both the proportion of participants who completed the intervention and their individual usage patterns. No reviews used subjective measures at all. Six reviews (9.8%) used usage logs as objective measures, capturing metrics such as frequency of logins, duration of time spent, recordings accessed, activity or homework completion, and self-reports. Another six reviews (9.8%) did not separate measurement of adherence from the concept of engagement. Five reviews

(8.2%) applied methods defined by the study authors, which were tailored to specific research objectives or to facilitate comparisons across studies. Examples included calculating rates or scores for meta-analysis (Witarto et al. 2022), percentage scores (Brown et al. 2016), grouping participants by adherence categories (Ellis, Draheim, and Anderson 2022; Winter et al. 2022), or providing direct comparisons across conditions (Forbes et al. 2023). One review (1.6%) measured adherence in combination with a related concept, specifically as a proxy for compliance (Moshe et al. 2021). Sample text used for each measurement category are shown in Table 5.

Table 4. Sample text for measuring engagement across included reviews.

Operationalisations of engagement	Example measurement
Completion metrics	"... the percentage of users who completed the entire intervention or completed the intervention per protocol, and/or the mean percentage of modules completed." (Valentine et al. 2025)
Usage metrics	"... frequency of use, indicators to capture milestones achieved, modules completed, number of logins." (Elkes et al. 2024)
Mixed measurement approaches	"... objective measures (e.g. usage pattern) and subjective measures (e.g. user experience)." (Ho et al. 2025)
Study-specific author-defined methods (<i>Ranking</i>)	"To assess engagement, we ranked the 21 studies in this review in percentile order in terms of intervention completion and intervention group study attrition." (Carolan, Harris, and Cavanagh 2017)
Measurement combined with adherence	"Engagement: user enrollment, attendance, session participation, homework completion, <i>adherence</i> , and dropout." (Bear et al. 2022)
Measurement combined with related concepts (<i>usability, satisfaction acceptability, feasibility</i>)	"Engagement: information on <i>usability, satisfaction, acceptability, or feasibility</i> of the app." (Anmella et al. 2022)

3.6. Factors influencing engagement and adherence

Across the 61 reviews, factors influencing engagement and adherence to Digital Mental Health Interventions (DMHIs) were reported at the participant, technology, and programme levels. These three categories emerged from inductively grouping factors into overarching themes, reflecting whether influences arose from the individual (participant), the tool itself (technology), or the broader delivery context (programme).

At the participant level, six studies (Borghouts et al. 2021; Brown et al. 2016; Simões de Almeida and Marques 2023; Garrido et al. 2019; Ho et al. 2025; Matis et al. 2020) noted that a user's demographic variables, specifically age, gender, employment, education, and housing situation, influenced their engagement. Five studies (Borghouts et al. 2021; Brown et al. 2016; Hanley and Wyatt 2021; Ho et al. 2025; Struthers et al. 2015) reported that ease of integration into everyday life influenced continued engagement, while four studies (Simões de Almeida and Marques 2023; Garrido et al. 2019; Matis et al. 2020; Struthers et al. 2015) noted the impact of pre-existing

Operationalisation of Adherence (by no. of studies)

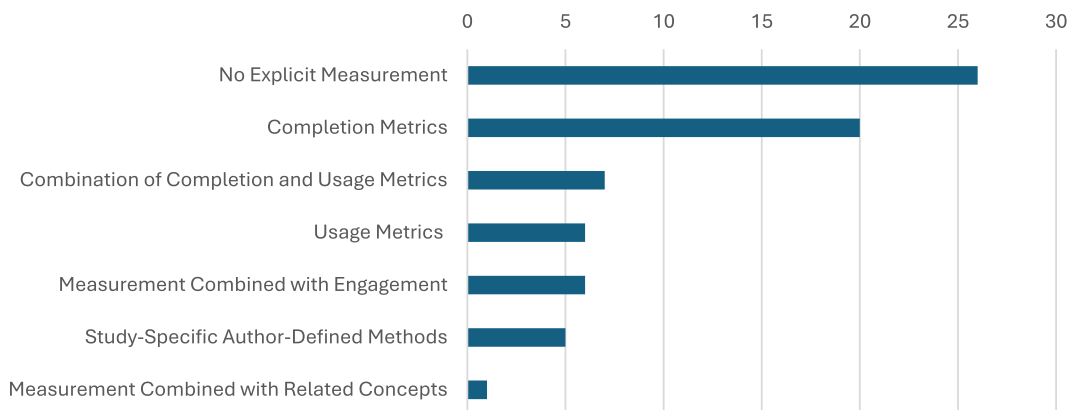


Figure 5. Summary of adherence measurements by number of studies.

mental health issues. Another four studies (Borghouts et al. 2021; Nunes-Zlotkowski et al. 2024; Struthers et al. 2015; Szinay et al. 2020) highlighted technological literacy, with high literacy described as a facilitator and low literacy as a barrier. Three studies also noted that particular personality traits affected an individual's decision to engage with a digital intervention (Borghouts et al. 2021; Hanley and Wyatt 2021; Ho et al. 2025).

At the technology level, six studies (Borghouts et al. 2021; Simões de Almeida and Marques 2023; Ho et al.

2025; Liverpool et al. 2020; Nunes-Zlotkowski et al. 2024; Oti and Pitt 2021) emphasised user's concerns regarding privacy, while five studies (Antezana et al. 2022; Simões de Almeida and Marques 2023; Nunes-Zlotkowski et al. 2024; Oti and Pitt 2021; Szinay et al. 2020) noted the importance of personalisation and customisation. Four studies (Antezana et al. 2022; Bear et al. 2024; Chan et al. 2022; Oti and Pitt 2021) examined user interface, defined as the degree of aesthetic quality, comfort, and functionality, and three studies identified technical issues (Brown et al. 2016; Hanley and Wyatt 2021; Nunes-Zlotkowski et al. 2024), referring to the tendency of the technology to malfunction.

At the programme level, perceived usefulness was reported in eight studies (Bear et al. 2024; Borghouts et al. 2021; Brown et al. 2016; Chan et al. 2022; Ho et al. 2025; Liverpool et al. 2020; Struthers et al. 2015; Szinay et al. 2020), and reliability and relevance of the DMHI content were noted in five studies (Borghouts et al. 2021; Simões de Almeida and Marques 2023; Hanley and Wyatt 2021; Ho et al. 2025; Oti and Pitt 2021). Two studies (Borghouts et al. 2021; Carolan, Harris,

Table 5. Sample text for measuring adherence across included reviews.

Operationalisations of adherence	Example measurement
Completion metrics	"Percentage of patients who completed all modules; number of finished modules per those prescribed; homework completion rates." (Chan et al. 2022)
Combination of completion and usage metrics	"... number of logins, time spent on the intervention, percentage who completed all activities/modules" (Linardon and Fuller-Tyszkiewicz 2020).
Usage metrics	"... app use data, including time spent and sessions using the app; characteristic adherence data through self-report" (Bear et al. 2022).
Measurement combined with engagement	"... quantified participant engagement using data such as dosage, exposure, or <i>adherence</i> ." (Davis et al. 2022)
Study-specific author-defined methods(<i>percentage score</i>)	"Percentage Score of Adherence Rate (to allow comparisons across interventions)." (Brown et al. 2016)
Measurement combined with related concepts (<i>compliance</i>)	"... the higher levels of <i>compliance</i> (often labelled as "adherence" in digital health research); Compliance is the proportion of participants completing assessments, the proportion of participants completing the first intervention module ..." (Moshe et al. 2021)

Table 6. Number of studies which discuss participant-, technology-, and programme-level factors influencing engagement and adherence.

Levels	Factors	No. of studies
Participant-level (n = 22)	Demographic variables	6
	Integration into everyday life	5
	Mental health status	4
	Technological literacy	4
	Personality traits	3
Technology-level (n = 18)	Privacy Concerns	6
	Personalisation	5
	User interface	4
	Technical issues	3
Programme-level (n = 15)	Perceived usefulness	8
	Content reliability and relevance	5
	Modalities	2

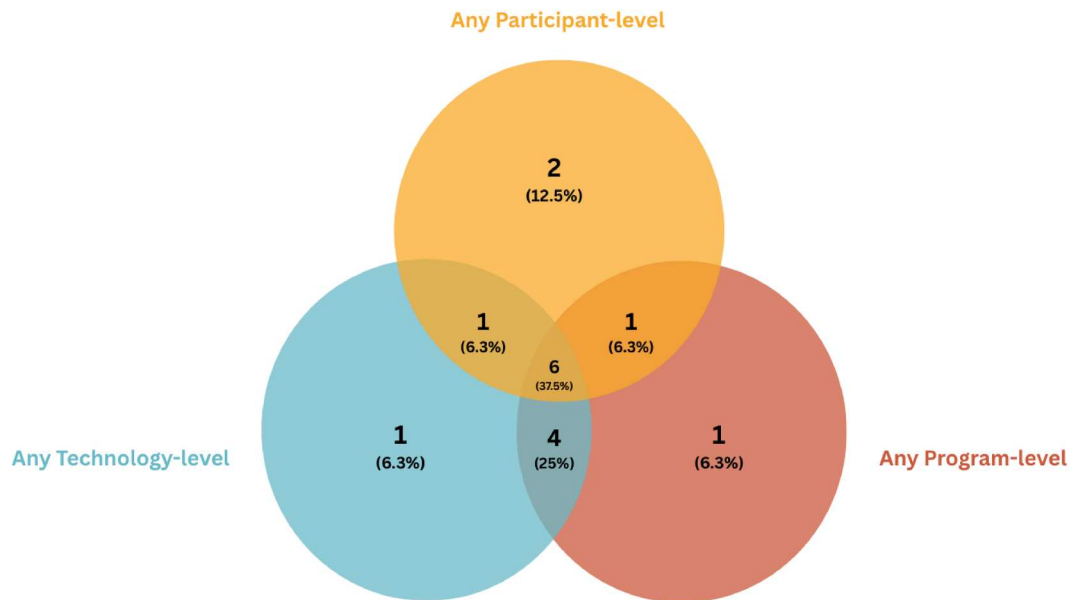


Figure 6. Venn diagram of overlapping studies examining factors affecting engagement and adherence.

and Cavanagh 2017) highlighted the role of delivery modalities, with more than one modality or delivery mode facilitating engagement. Six studies did not discuss the factors affecting engagement and adherence to DMHIs. A summary of these factors and their corresponding number of studies is presented in Table 6.

Because the themes were developed inductively from the data, a single study could contribute to multiple factors and appear across more than one level. Further details on the thematic synthesis and specific study overlaps are provided in Supplementary Material 3. Of the included studies, six (37.5%) discussed factors at all three levels. Four studies (25%) addressed both technology- and programme-level factors, one study (6.3%) covered both person- and technology-level factors, and one study examined both person- and programme-level factors. Two studies (12.5%) focused solely on person-level factors, while one study (6.3%) each focused solely on technology-level and programme-level factors. Figure 6 illustrates these overlaps among studies examining factors influencing engagement and adherence.

4. Discussion

This umbrella review synthesised 61 systematic, scoping, meta-analytic, and integrative reviews to examine how engagement and adherence are conceptualised, measured, and influenced in digital mental health interventions (DMHIs). Across diverse delivery methods, populations, and outcomes, review characteristics and quality were assessed to gauge evidence strength. Despite heterogeneity, the synthesis revealed consistent

findings, patterns, and research gaps, with implications for reporting DMHI studies and designing DMHIs.

The three most frequently reported definitions of engagement across the included studies characterised it as the extent of programme use and completion (11.5%), as a multi-faceted experience (9.8%), and as initial uptake and continued use (6.6%). The first and third definitions lend themselves to quantitative measurement (e.g. usage and completion metrics), whereas the second is more appropriately captured through qualitative approaches that reflect the experiential aspects of engagement. Taken together, these suggest that engagement reflects *the extent to which a user interacts with the intervention, and is a multi-faceted and dynamic process of initial uptake and sustained use*.

Nevertheless, operationalisation in practice tends to rely heavily on quantitative usage indicators, such as login frequency, time spent, or completion of modules. Some reviews, however, proposed more nuanced approaches. For instance, Fleming et al. (2018) distinguished three levels of engagement: minimal use (initiation of the intervention), moderate use (greater than minimal but short of sustained), and sustained use (completion of the intervention). Other methods were tailored to particular delivery modalities, such as video games (Oti and Pitt 2021) or social networking platforms (Ridout and Campbell 2018). One review also applied the CAPE Model (Connect–Attendance–Participation–Enact), which offers a theoretical framework for examining engagement across pre-use, active use, and post-use phases of DMHIs (Davis et al. 2022).

Compared to engagement, definitions of adherence were considerably more consistent across reviews, with the most common being treatment completion (16.4%) and usage as prescribed or intended (13.1%). Synthesised, this indicates that adherence refers to *using an intervention as intended by the study authors, which typically involves completing programmes, modules, sessions, treatment protocols, or meeting a minimum prescribed dosage*. As a result, adherence is usually assessed quantitatively through rates or percentages of completion, objective metrics from the DMHI (such as frequency of logins, duration of use, activity or homework completion, and self-reports), or a combination of both. Like engagement, several reviews introduced unique measurement approaches tailored to their research objectives, including calculating group rates or scores to facilitate comparison across studies. For instance, Witarto and colleagues (2022) conducted a meta-analysis, while Brown et al. (2016) developed percentage-based adherence scores. Some reviews also categorised participant adherence into groups (Ellis, Draheim, and Anderson 2022; Winter et al. 2022).

A key challenge in this domain is that engagement and adherence often overlap, with majority of the included studies failing to clearly distinguish between them. Indeed, engagement and adherence are often used interchangeably with terms like acceptability, usability, feasibility, retention, and attrition (Anmella et al. 2022; Borghouts et al. 2021; Ellis, Draheim, and Anderson 2022; Polak and Tanzer 2024). Nonetheless, differences were evident: engagement definitions emphasised the extent and nature of use, while adherence referred to using the intervention as intended. Engagement also uniquely encompassed initial uptake and continued use, processes not typically associated with adherence. Delineating these two constructs allow for more meaningful comparisons across research findings, and contributes to the development of standardised evaluation frameworks. The main goal is to design studies that address this overlap, a task that is more complex than it may first appear.

Taken together, the literature suggests that engagement is best conceptualised as an overarching process that spans before, during, and after interaction with a DMHI. Before interaction refers to users' motivations and intentions to engage; during interaction captures the nature and intensity of use; and after interaction reflects sustained application and long-term outcomes. This is seen within the CAPE Model (Connect–Attend–Participation–Enact) theoretical framework for examining engagement, which discusses pre-use, active use, and post-use phases of DMHIs (Davis et al. 2022). Zainal and colleagues (2025) similarly categorised

engagement as treatment initiation (before), extent of DMHI usage (during), and module completion (after).

These frameworks distinguish engagement from adherence by emphasising its conceptual focus on the quality and depth of interaction. This conceptual distinction brings a major challenge for measurement. As noted earlier, module usage and completion metrics are applied to both engagement and adherence. For adherence, it can be clearly seen how these metrics could be used to indicate whether an individual's use matches the recommended protocol. However, it remains unclear whether module usage and completion metrics adequately capture engagement as a process occurring before, during, and after interaction with a DMHI. This underscores the measurement challenge of integrating subjective feedback measures alongside objective metrics to fully capture the quality and depth of user engagement.

Notably, the small number of studies that assessed publication bias raises concerns that the literature may disproportionately report positive associations between engagement and outcomes, highlighting that, discontinued use may not necessarily reflect a negative outcome. In some reviews, participants reportedly 'felt better' after only a few modules, accounting for disengagement (Brown et al. 2016; Nunes-Zlotkowski et al. 2024). Engagement is not an instantaneous event, but rather unfolds through ongoing decision-making, with users cycling through phases of engagement, disengagement, and reengagement over time (Simões de Almeida and Marques 2023; Wähnke et al. 2024). There is a call to reframe disengagement as a natural part of the usage cycle, as this can be interpreted as users having achieved their recovery goals. The challenge lies in how to capture the distinct elements of each, both in terms of definition and measurement.

This measurement challenge aligns with decade-long calls to move beyond blunt indicators of compliance. Struthers et al. (2015) highlighted the shortage of rigorous qualitative research on DMHI engagement, noting that this gap hinders a deeper understanding of user experiences. Effective models of engagement will need subjective feedback to contextualise objective measures (Molloy and Anderson 2021). Despite ongoing calls to supplement objective measures with more subjective assessments, most studies continue to rely on metrics such as module completion rates and frequency of use to define and measure engagement and adherence. This is because it raises several practical and methodological considerations. For example, what is the feasibility of collecting and analysing qualitative feedback from DMHIs deployed at scale and used frequently by individual users? What is the feasibility of collecting

qualitative data that is open enough to capture individual experiences across diverse DMHIs (e.g. gratitude diaries versus mindfulness interventions), yet structured enough to allow standardised comparisons of users' experiences across interventions?

Nevertheless, current methods for capturing and reporting engagement metrics often fail to fully reflect their dynamic process. In conceptualising and operationalising engagement, researchers can move beyond simple binary measures and de-emphasise intervention completion, instead consider longer-term usage patterns that incorporate meaningful disengagement. Such studies can help clarify when disengagement represents a deliberate, positive choice, such as 'feeling better', rather than a negative outcome. Although rigorous qualitative work remains limited, subjective methods like interviews, focus groups, think-aloud protocols, and written reports (Bijkerk et al. 2023) offer valuable insights into user experiences. Further research is required to determine effective ways to integrate subjective, user-centered data into large-scale DMHI research.

Another approach to addressing the 'engagement paradox' – that is, disengagement once participants have reached their goals – is the concept of optimal engagement. Badawi et al. (2023) define the 'optimal dose' as the level of use required for a digital intervention to achieve its intended therapeutic effect. Building on this, Boucher and Raiker (2024) recommend a 'minimally effective dose' (14) delivered within the shortest possible timeframe. Such single-session interventions have shown promise in being both effective and acceptable to young people. Smith and colleagues (2025) extend this perspective by encouraging digital mental health intervention (DMHI) researchers to determine the 'target dose' (4) for their interventions. This is the engagement threshold most likely to maximise both uptake and clinical outcomes.

The concept of an 'optimal dose' is further complicated by variability in engagement and adherence, particularly in relation to individual differences and treatment modality. Similar to offline mental health interventions, the nature and severity of a user's presenting symptoms may influence adherence to digital treatments (Garrido et al. 2019; Matis et al. 2020). For instance, a user seeking short-term support for an acute anxiety episode may engage differently from someone requiring ongoing support for generalised anxiety disorder. Variability in treatment modality also shapes how engagement and adherence are conceptualised and measured. For example, metrics such as 'number of pages viewed' may be appropriate for psychoeducational interventions but are not applicable to

modalities such as mindfulness-based exercises. Additionally, the use of multiple treatment modalities has been shown to influence engagement with digital mental health interventions (Borghouts et al. 2021; Gan et al. 2021).

Therefore, it is important to move beyond a one-size-fits-all approach to defining and measuring engagement and adherence and acknowledge inherent conceptual and operational variability. This then leads into a discussion on what factors influence engagement and adherence.

4.1. Factors influencing engagement and adherence

The third research question of this review examines factors influencing engagement with DMHIs, which can be conceptualised across three clusters of factors: (i) participant-level factors, (ii) technology-related factors, and (iii) programme level factors. Participant-level characteristics include pre-existing user attributes and experiences that shape both the decision to adopt a DMHI and the likelihood of sustained engagement. Personality traits have been identified as one such factor, with evidence suggesting that neuroticism, agreeableness, openness, and resistance to change are associated with greater engagement, whereas extraversion may be linked to lower engagement (Borghouts et al. 2021). A user's prior experience with technology and their level of technical literacy also play important roles: high levels in these areas facilitate engagement, whereas low levels can hinder it (Nunes-Zlotkowski et al. 2024; Szinay et al. 2020). Mental health status is another key factor. Higher baseline depression scores and a history of mood disorders are linked to greater adherence, while elevated anxiety scores have been found to predict increased dropout rates (Garrido et al. 2019; Matis et al. 2020; Struthers et al. 2015).

Technology-level factors pertain to features inherent to the digital delivery of interventions, which can either facilitate or impede engagement depending on how effectively they create a seamless and trustworthy user experience. As technology becomes increasingly central to daily life, issues such as data security and technical malfunctions were frequently highlighted across reviews. Trust in technology strongly influences engagement, with higher perceived trust correlating with greater intentions to use DMHIs (Sawrikar and Mote 2022). Users often experience a 'privacy paradox', where they engage with DMHIs despite mistrust, driven by convenience and lack of alternatives (Gentry et al. 2024). Providing clear information about data handling practices can help build user confidence and promote

engagement (Martinez-Martin 2020). In addition, user engagement was influenced both by the ability to personalise the intervention according to individual preferences and needs and by perceptions of its aesthetics, comfort, and overall functionality. This aligns with established principles of user interface and user experience (UI/UX) design (Alqahtani and Orji 2020).

Factors influencing user engagement and adherence are also present in the programme-level, particularly the structure, content, and delivery of DMHIs. Content relevance and reliability are intricately linked to perceived usefulness; participants are more likely to engage if they view the intervention as a good fit and find the content both credible and relevant (Borghouts et al. 2021). Notably, perceived usefulness emerged as the most frequently cited facilitating factor, identified in eight reviews. Perceived usefulness refers to the extent to which people find a technology to facilitate their goals – which, in the context of digital health, typically refers to symptom improvement. This construct is linked to perceived ease of use, or the degree to which users feel the intervention will be straightforward and accessible (Park and Kim 2023). Research demonstrates that perceived usefulness is moderately associated with stronger intentions to use DMHIs, greater participant satisfaction, and improved therapeutic alliance (Huffman et al. 2023; Sawrikar and Mote 2022).

In response to these determinants of engagement and adherence, various strategies have been implemented in DMHI studies. Reminders and notifications, for instance, help reduce disengagement by prompting users to interact with interventions at key moments and facilitating their integration into daily routines (Simões de Almeida and Marques 2023; Elkes et al. 2024; Linardon and Fuller-Tyszkiewicz 2020; Winter et al. 2022). For specific demographic groups, the inclusion of social features such as peer support, group challenges, or opportunities for community interaction has been shown to positively influence engagement and adherence (Biagianti, Quraishi, and Schlosser 2018; Ridout and Campbell 2018; Yap and Lin 2025; Zainal et al. 2025). To address technology-level concerns related to user interface design and personalisation, strategies like gamification (Antezana et al. 2022; Bear et al. 2024; Brown et al. 2016; Simões de Almeida and Marques 2023; Szinay et al. 2020) and persuasive technologies (Carolan, Harris, and Cavanagh 2017; Wildeboer, Kelders, and van Gemert-Pijnen 2016) have also proven effective.

Notably, onboarding processes such as providing training and tutorials were the only strategy directly linked to technology acceptance and perceived usefulness, despite these constructs being consistently identified as central to engagement (Borghouts et al. 2021;

Nunes-Zlotkowski et al. 2024; Szinay et al. 2020). Allowing users to have a say in their user experience underscores the importance of co-production and user-centered design, particularly given that expectations and approaches to technology shift considerably across generations and over time (Boucher and Raiker 2024; Lipschitz et al. 2023; Sawrikar and Mote 2022).

5. Strengths and limitations

To the best of our knowledge, this umbrella review is the first to systematically synthesise systematic reviews and meta-analyses that define and measure engagement and adherence in digital interventions for mental health. This review offers several notable strengths. All included reviews are both timely and relevant. The generally high quality of the included reviews further strengthens confidence in the findings. Methodological rigour was maintained by adhering to JBI guidelines, including critical appraisal, data extraction, and meta-aggregation, with any deviations from the published protocol clearly recorded and justified. Publishing the protocol in advance and making both the search strategy and data synthesis process readily available enhances the transparency and replicability of the review.

Several limitations should also be noted. A citation matrix or corrected covered area could not be conducted, so the proportion of overlap in primary studies was not directly quantified. However, overlap by topic and frequency was reported wherever possible, and details on the critical appraisal scores and assessment tools used in the included reviews were provided. The meta-aggregation process was conducted solely by the primary author, which may introduce potential bias. Some relevant studies may have been excluded due to the omission of gray literature; to address this, forward citation chaining was performed. The limited number of studies from Asia may reflect the English-language inclusion criteria, suggesting that research from non-English-speaking regions is likely underrepresented. This is consistent with trends reported in other systematic reviews (Elkes et al. 2024). Most included studies focused on common mood and anxiety disorders, with fewer addressing more severe or less frequently studied mental health conditions. Expanding search queries or publication dates could help identify additional relevant evidence. The review also focused on digital mental health specifically, and future research can look into defining and measuring engagement and adherence in the digital health domain more generally.

Finally, it is important to note that the reported findings on definitions and measurements are based

on the proportion of the total 61 studies, whereas not all articles addressed all elements. This means it may be the case that a study had a measurement of adherence without defining it, but it may also be the case that a study did not address adherence at all and rather focused on engagement. To guide interpretation, Supplementary Material 6 provides details on whether each study reported a definition and/or measurement of engagement and adherence.

6. Conclusion and recommendations

This umbrella review examines the varied definitions and measurements of engagement and adherence in digital mental health interventions. Despite the inherent variability in conceptualisation and operationalisation, reporting, design, and evaluation practices can still be structured to allow for transparency, standardisation, and meaningful comparisons across studies. By meta-aggregating and synthesising 61 reviews, it lays the groundwork for more consistent evaluation, facilitating cross-study comparisons and supporting the development of robust evidence bases.

The study presents recommended practices to address specific challenges identified in the review and offer guidance moving forward. This is a call for transparency and consistency across reporting and design and to provide explicit definitions of engagement and adherence, outline the chosen measures and their foundations, and justify how these measures are implemented and why they are relevant. Relevant challenges are outlined within Table 7, with Table 7 also

presenting recommendations that are relevant for reporting of DMHIs.

Table 8 outlines recommendations for the design of DMHIs. Together, these tables encourage thoughtful consideration in the design and reporting of DMHI studies while recognising that not all practices will apply to every study.

The literature agrees that engagement is an overarching, multidimensional process involving user interaction, but current measurement methods do not fully capture this complexity. Advanced longitudinal approaches are encouraged to move beyond binary outcomes and to recognise disengagement as a dynamic process. Simultaneously, deeper qualitative inquiry can enrich our understanding of user experiences. Adherence is typically defined as intervention completion, yet its definition varies across modalities; establishing clearer reporting standards and standardised adherence scores would enable more meaningful comparisons. Discussions also addressed what an optimal dose might look like, how it could be determined, and how this might inform the definition and measurement of adherence. Another area highlighted for consideration was how disengagement might be differentiated from successful completion.

Both engagement and adherence are influenced by participant, technology, and programme-level factors, underscoring the need for strategies that enhance users' acceptance and perceived usefulness of technology. Ultimately, meaningful co-production among users, researchers, and technology developers is crucial. This collaboration ensures that digital interventions are grounded in lived experience, informed by scientific rigour, and supported by technical feasibility.

Overall, DMHIs offer great potential in terms of leveraging technology to provide psychological support, information, and therapy through various platforms. Although these are scalable and accessible, low engagement and adherence currently limit their effectiveness. If efforts to promote engagement and adherence are to

Table 7. Recommended practices for reporting of DMHIs to address specific challenges.

Challenge	Reporting DMHI Studies
Adherence and engagement overlap in some respects but also differ.	To facilitate cross-study comparisons, methodology sections should clearly outline how researchers have defined and measured engagement and adherence.
Similar metrics are often applied to both engagement and adherence, but these may not completely capture their unique elements.	Reporters should consider the user metrics available to them and be transparent regarding the caveats of these as true indicators of engagement and adherence.
Inconsistent reporting limits understanding of how adherence and engagement link to symptom improvement.	Where DMHIs are trialled, there should be preregistration of transparent protocols which outline how adherence and engagement will be conceptualised and measured, with any deviations from the protocol clearly noted within actual results.
Disengagement may occur for a variety of reasons, from lack of user satisfaction to symptom improvement.	Reporters should track and report reasons for disengagement, whether positive or negative.

Table 8. Recommended practices for the design of DMHIs.

Design of DMHIs
In the design of DMHIs, stakeholders should consider the context of the use in terms of measures of engagement and adherence that are appropriate to the population for whom the intervention is being designed.
The design of the intervention should include calculation of its minimally effective 'dose' using measures of engagement and adherence.
When measures of engagement and adherence have been selected, they should be validated in terms of establishing the most effective and appropriate level of engagement and adherence to help identify symptoms and to promote symptom improvement.
Intervention design should offer opportunities to the user of the DMHI to report on reasons behind nonadherence and disengagement.

be effective, a key starting point is establishing clear definitions or a theoretical framework to guide their understanding, as well as guidelines for their measurement and reporting.

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Author contributions

CRedit: **Lyen Krenz Yap**: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft; **Edel Ennis**: Conceptualization, Project administration, Supervision, Writing – review & editing; **Maurice Mulvenna**: Conceptualization, Supervision, Writing – review & editing; **Jorge Martinez**: Conceptualization, Project administration, Supervision.

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